

Year 7 · Chapter 6 · Geometrical figures: answers (teacher copy)

6.01 Transforming shapes (booklet p. 3)

WE DO examples

- Ex 1: mirror image on the right of the line
- Ex 2: moved 5 right and 3 down
- Ex 3: turned upside down through O
- Ex 4: turned a quarter turn clockwise
- Ex 5: mirror image below the line
- Ex 6: turned three-quarters clockwise

Set A

- reflected to the right of $x = 7$
- reflected below the line

Set B

- moved 4 left, 1 up
- moved 4 right, 6 up

Set C

- rotated 180°
- rotated 90° clockwise

Set D

- rotated 270° clockwise
- reflected in the diagonal

Extension

- a rotation of 180° about the point where the lines cross
- Yes: both end in the same position

Exit ticket

- Level 1: reflection
- Level 2: (6, 2)
- Level 3: it points right

6.02 Combined transformations (booklet p. 7)

WE DO examples

- Ex 1: translated, then flipped over the line
- Ex 2: rotated, then moved 3 left
- Ex 3: reflected, then moved down 1
- Ex 4: Often yes: the images end in different places
- Ex 5: three images drawn in order
- Ex 6: e.g. rotate 180° , then translate

Set A

- moved then flipped
- flipped then moved up

Set B

- turned then moved down
- moved then turned

Set C

- three images
- three images

Set D

- e.g. rotate 180° then translate
- reflect in a vertical line; or reflect then translate
- Not always: e.g. a reflection in a horizontal line and a translation to the right give the same, but a translation up does not
- translations

Extension

- (-1, -5)
- (-4, -1)

Exit ticket

- Level 1: the image after one transformation
- Level 2: (4, 6)
- Level 3: (0, -3)

6.03 Line symmetry (booklet p. 11)

WE DO examples

- Ex 1: 2
- Ex 2: 4
- Ex 3: 6
- Ex 4: 0
- Ex 5: 2
- Ex 6: infinitely many

Set A

- 3
- 1
- 2
- 1
- 1
- 0

Set B

- 5
- 4
- 1

Set C

- 8
- 0
- 0
- 1
- 0
- 2

Set D

- 10
- e.g. A, B, C, D, E, M, T, U, V, W, Y
- e.g. a rectangle
- No: 2 axes would make all sides equal, giving 3

Extension

- a shape with 2 axes
- 12

Exit ticket

- Level 1: 3
- Level 2: 1
- Level 3: 6

6.04 Rotational symmetry (booklet p. 15)

WE DO examples

- Ex 1: 2
- Ex 2: 4
- Ex 3: 2
- Ex 4: none (order 1)
- Ex 5: 2
- Ex 6: 72°

Set A

1. order 3
2. order 2
3. none
4. order 6
5. none
6. order 4

Set B

1. 2
2. 5
3. none

Set C

1. 8, 45°
2. 4, 90°
3. 2, 180°
4. 2, 180°
5. 3, 120°
6. none

Set D

1. e.g. H, I, N, O, S, X, Z
2. Yes, e.g. a parallelogram or the letter S
3. 5
4. e.g. a square or a plus sign

Extension

1. 12
2. order 6: 60° is larger than 45°

Exit ticket

1. Level 1: 4
2. Level 2: no
3. Level 3: 60°

6.05 Classifying triangles (booklet p. 19)**WE DO examples**

1. Ex 1: equilateral, acute-angled
2. Ex 2: scalene, right-angled
3. Ex 3: isosceles, acute-angled; $\angle F = \angle H$
4. Ex 4: scalene, obtuse-angled
5. Ex 5: scalene, acute-angled
6. Ex 6: equilateral, acute-angled

Set A

1. isosceles
2. scalene
3. equilateral
4. isosceles
5. scalene
6. isosceles

Set B

1. right-angled
2. acute-angled
3. obtuse-angled

Set C

1. isosceles, right-angled
2. isosceles, obtuse-angled
3. equilateral, acute-angled
4. scalene, right-angled
5. scalene, acute-angled
6. isosceles, acute-angled

Set D

1. scalene, acute-angled
2. Yes: isosceles right-angled (the 7 cm side is the longest)
3. isosceles, obtuse-angled
4. No: equilateral angles are all 60°

Extension

1. 40° and 40°
2. equilateral

Exit ticket

1. Level 1: equilateral
2. Level 2: isosceles, right-angled
3. Level 3: isosceles, obtuse-angled

6.06 Angle sum of a triangle (booklet p. 23)**WE DO examples**

1. Ex 1: $x = 59$
2. Ex 2: $u = 51$
3. Ex 3: $a = 127$
4. Ex 4: $y = 50, z = 80$
5. Ex 5: $n = 62, m = 56$
6. Ex 6: $\angle PQR = 36^\circ$ (C)

Set A

1. $x = 59$
2. $v = 76$
3. $k = 50$
4. $d = 85$
5. $u = 51$
6. $p = 55$

Set B

1. $a = 127$
2. $p = 100$
3. $w = 27$

Set C

1. $y = 50, z = 80$
2. $n = 62, m = 56$
3. $t = 60$
4. $b = 35$
5. $c = 45$
6. $q = 36$

Set D

1. $x = 30$
2. 72° each
3. No: $100 + 90$ is already more than 180
4. $53\frac{1}{3}^\circ, 53\frac{1}{3}^\circ, 73\frac{1}{3}^\circ$

Extension

1. they form a straight line (180°)
2. two obtuse angles already add to more than 180°

Exit ticket

1. Level 1: 70°
2. Level 2: 62°
3. Level 3: 46°

6.07 Exterior angle of a triangle (booklet p.

27)

WE DO examples

1. Ex 1: $x = 115$
2. Ex 2: $x = 111$
3. Ex 3: $x = 112$
4. Ex 4: $p = 96$
5. Ex 5: $m = 99$, $n = 81$
6. Ex 6: it is the sum of both, so it is bigger than either

Set A

1. $x = 115$
2. $x = 111$
3. $a = 110$
4. $b = 120$
5. $c = 120$
6. $d = 90$

Set B

1. $x = 70$
2. $p = 96$
3. $y = 55$

Set C

1. $m = 99$, $n = 81$
2. $j = 108$, $k = 72$
3. $r = 55$, $s = 70$
4. $t = 60$
5. $u = 62$, $v = 90$
6. $w = 60$

Set D

1. $x = 37$
2. 70° , 70° , 40°
3. 360°
4. Yes, if the adjacent interior angle is obtuse

Extension

1. 144°
2. 55° and 75°

Exit ticket

1. Level 1: 115°
2. Level 2: 65°
3. Level 3: 40°

6.08 Quadrilaterals (booklet p. 31)

WE DO examples

1. Ex 1: non-convex
2. Ex 2: parallelogram
3. Ex 3: rhombus
4. Ex 4: trapezium
5. Ex 5: a parallelogram with all sides equal
6. Ex 6: trapezium

Set A

1. convex
2. non-convex
3. convex
4. non-convex
5. convex
6. non-convex

Set B

1. rectangle
2. kite
3. square

Set C

1. 2 axes, order 2
2. 1 axis, none
3. 4 axes, order 4
4. 0 axes, order 2
5. 2 axes, order 2
6. 1 axis, none

Set D

1. parallelogram
2. a parallelogram with a right angle
3. Yes: it is a parallelogram with all sides equal
4. Yes: both pairs of opposite sides are parallel

Extension

1. a kite is separate; yes, a rhombus is a kite
2. rectangle and rhombus

Exit ticket

1. Level 1: yes
2. Level 2: trapezium
3. Level 3: a quadrilateral with both pairs of opposite sides parallel

6.09 Angle sum of a quadrilateral (booklet p.

35)

WE DO examples

1. Ex 1: $y = 70$
2. Ex 2: $x = 81$
3. Ex 3: $m = 26$
4. Ex 4: $y = 59$
5. Ex 5: a diagonal splits it into 2 triangles: $2 \times 180^\circ$
6. Ex 6: 90° each

Set A

1. $y = 70$
2. $x = 81$
3. $m = 26$
4. $a = 105$
5. $b = 95$
6. $c = 55$

Set B

1. $y = 59$
2. $d = 63$, $e = 117$
3. $f = 120$

Set C

1. $x = 53\frac{1}{3}$
2. $k = 60$
3. $p = 110$, $q = 70$, $r = 110$
4. $t = 95$
5. $u = 87$
6. $w = 36$

Set D

1. 36° , 72° , 108° , 144°
2. No: $4 \times$ more than 90° is more than 360°
3. 122° , 58° , 122°
4. 90° each

Extension

1. 720°
2. 108°

Exit ticket

1. Level 1: 110°
2. Level 2: 80°
3. Level 3: 115° , 65° , 115°

6.10 Properties of quadrilaterals (booklet p. 39)**WE DO examples**

1. Ex 1: e.g. opposite sides equal; opposite angles equal; diagonals bisect each other
2. Ex 2: C
3. Ex 3: rectangle, square
4. Ex 4: rectangle, square
5. Ex 5: parallelogram, rectangle, rhombus, square
6. Ex 6: parallelogram, rectangle, rhombus, square

Set A

1. true
2. false
3. true
4. false
5. true
6. false

Set B

1. rhombus, square
2. rectangle, square
3. parallelogram, rhombus, rectangle, square
4. rhombus, square, kite
5. trapezium
6. rhombus, square

Set C

1. bisect each other
2. equal, bisect
3. bisect at 90°
4. equal, bisect at 90°
5. one bisects the other at 90°
6. none of these (equal if isosceles)

Set D

1. $CD = 7$ cm, $AD = 4$ cm, $\angle C = 65^\circ$
2. 90°
3. square
4. kite

Extension

1. 5 cm
2. the diagonal bisects each 90° angle

Exit ticket

1. Level 1: rectangle or square
2. Level 2: they bisect each other at right angles
3. Level 3: rectangle (or square)